



Full length article

Prenatal exposure to prescribed smoke PM_{2.5} and low birth weight in Georgia, USA

Lei Li^a, Hua Hao^a, Jennifer D. Stowell^b, Emily M. Gargulinski^c, Kamal J. Maji^d, Colleen E. Reid^e, Talat M. Odman^f, Howard H. Chang^g, Amber Soja^h, Yang Liu^{a,*}

^a Department of Environmental Health, Rollins School of Public Health, Emory University, Atlanta, GA 30322, USA

^b Department of Environmental Health, Boston University School of Public Health, Boston, MA 02118, USA

^c National Institute of Aerospace, Hampton, VA 23666, USA

^d Department of Public Health, Environments and Society, London School of Hygiene and Tropical Medicine, London, UK

^e Department of Geography, University of Colorado Boulder, Boulder, CO 80309, USA

^f Georgia Institute of Technology, Atlanta, GA 30332, USA

^g Department of Biostatistics and Bioinformatics, Rollins School of Public Health, Emory University, Atlanta, GA 30322, USA

^h NASA Langley Research Center, Hampton, VA 23681, USA

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ABSTRACT

Background: Prescribed burning is a major but poorly characterized source of fine particulate matter (PM_{2.5}) in the southeastern United States. While wildfire-related smoke PM_{2.5} has been linked to adverse birth outcomes, evidence on prescribed fire-related smoke (PFS) and birth weight remains limited, particularly in the Southeast. **Objective:** This study investigated the association between prenatal exposure to PFS-specific PM_{2.5} and birth weight in Georgia, USA.

Methods: We analyzed 448,477 singleton term births (37–42 weeks of gestation) with last menstrual period (LMP) dates between January 1, 2013 and December 31, 2016. Daily ZIP code-level PFS PM_{2.5} concentrations were averaged over early gestation (first 4 and 6 weeks), each trimester, and the entire pregnancy. High-exposure metrics were defined as the number of days exceeding percentile-based thresholds for each exposure window. Linear regression models adjusted for non-PFS PM_{2.5}, LMP date, gestational age, infant sex, and maternal characteristics.

Results: Higher PFS PM_{2.5} exposure was independently associated with lower birth weight. Each 1 µg/m³ increase in average PFS PM_{2.5} during the first 4 weeks and the first trimester was associated with birth weight decreases of 3.22 g (95 % confidence interval [CI]: 0.82, 5.62) and 4.56 g (95 % CI: 0.12, 9.00), respectively. Each additional day exceeding the 95th percentile (1.21 µg/m³) of PFS PM_{2.5} during the entire gestation corresponded to an 11.70 g (95 % CI: 0.95, 22.45) reduction.

Conclusion: Prenatal PFS exposure may reduce birth weight, particularly during early pregnancy, and both exposure intensity and duration are important in evaluating the health impacts of fire-related PM_{2.5}.

1. Introduction

Wildland fires, including both unplanned wildfires and managed prescribed burns, are a major source of outdoor air pollution and an important contributor to seasonal increases in fine particulate matter (PM_{2.5}). In the United States, recent analyses indicate that wildland fires account for more than 30 % of primary PM_{2.5} emissions nationwide (Jaffe, 2020). Prescribed burning is a necessary land management practice that applies fire under defined conditions to maintain fire-

dependent ecosystems, reduce fuel accumulation, and lower the risk of large and uncontrolled wildfires (Kolden, 2019; Collins, 2023). In the southeastern United States; prescribed fire is widely used as a routine management tool and contributes substantially to regional seasonal smoke burdens. This region issues approximately 70 % of all prescribed fire permits nationally (Kolden, 2019), and prescribed burns are responsible for about 80 % of all fire-related PM_{2.5} emissions in this area (Jaffe, 2020). During the peak burning season from January through April; prescribed fires can contribute approximately 20 % of daily

* Corresponding author.

E-mail address: yang.liu@emory.edu (Y. Liu).

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ambient PM_{2.5} concentrations in Georgia and surrounding areas (Maji, 2024).

Prescribed fires have received increasing attention in recent years due to their expanding use and potential implications for public health. Although prescribed burning is widely recognized as both desirable and necessary on fire-adapted landscapes, as reflected in recent national assessments of wildland fire management (Commission, W. F. M. a. M. ON FIRE: The Report of the Wildland Fire Mitigation and Management Commission, (U.S. Department of Agriculture, 2023); smoke emissions from prescribed fires can influence population exposure. In the southeastern United States, prescribed burning frequently occurs at the interface of wildland or agricultural land and urban areas, where smoke may affect nearby communities on a recurring basis (Williamson et al., 2012; Li, 2023). Although pollutant concentrations during prescribed burns are typically lower than those observed during large wildfire events, the higher frequency and closer proximity of these exposures raise important public health considerations.

Unlike wildfires, prescribed burns are carefully planned and regulated through permits and management guidelines, making them a controllable and policy-driven source of smoke exposure (Ryan et al., 2013). As a result, the health impacts of prescribed fire smoke (PFS) may be mitigated through informed public health strategies and individual actions, such as minimizing outdoor activities during smoke events, improving indoor air filtration, expanding access to HEPA filters in homes and schools, and enhancing risk communication (Jaffe, 2020; Haikerwal, 2015). Understanding the health effects of PFS is therefore essential for informing these interventions. Integrating health-relevant evidence into prescribed fire decision-making could help balance the ecological benefits of fire with the need to minimize smoke-related harms, underscoring the importance of interdisciplinary research that links fire management, air quality, and public health.

Prescribed fires and wildfires also differ in combustion conditions and therefore in smoke composition. Prescribed burns are typically conducted under cooler and moister conditions and often consume finer fuels such as grasses and understory vegetation. These combustion regimes favor increased smoldering processes that produce particulate matter with higher fractions of organic carbon and semi-volatile organics (Jaffe, 2020; Kelp, 2023). In contrast, wildfires generally burn under hotter and drier conditions, generating more intense flaming combustion, greater quantities of black carbon, and distinct profiles of gaseous pollutants. Wildfires also emit high-temperature combustion by-products, including greater amounts of polycyclic aromatic hydrocarbons (PAHs), such as benzo[a]pyrene (Jaffe, 2020; Kiely, 2024). Field and modeling studies suggest that fires conducted under prescribed burn conditions emit less PM_{2.5} and lower amounts of certain combustion gases per unit of fuel burned than uncontrolled wildfires, although emission magnitudes vary by setting and fuel type (Urbanski, 2014). These compositional and emission differences imply the potential for varying health effects per unit mass of PM_{2.5}; yet epidemiologic evidence that distinguishes PFS-specific PM_{2.5} from wildfire and other PM_{2.5} sources remains limited (Long et al., 2022; Stowell, 2025).

Birth weight is a key indicator of neonatal health and long-term developmental potential. Low birth weight is associated with increased risks of infant morbidity and mortality, as well as metabolic and cardiovascular diseases later in life (Wang, 2022; Wang, 2022; Kavousi and Snieder, 2024; Saigal and Doyle, 2008). Numerous epidemiological studies have shown that maternal exposure to ambient PM_{2.5} during pregnancy is associated with reduced birth weight and other adverse perinatal outcomes (Fang, 2020; Xiao, 2018; Strickland, 2019). A growing body of literature has also examined prenatal exposure to PM_{2.5} from landscape fire smoke, particularly wildfire smoke (WFS), in relation to birth weight and preterm birth (Zhang, 2023; Fernandez, 2023; Abdo, 2019; Holstius et al., 2012; Li, 2021; O'Donnell and Behie, 2015; Gao, 2026). However, the evidence on WFS and birth weight remains mixed. While several studies have reported lower birth weight following WFS exposure (Zhang, 2023; Abdo, 2019; Holstius et al.,

2012; Li, 2021), others have observed higher birth weight, macrosomia, or increased odds of large-for-gestational-age birth (Fernandez, 2023; O'Donnell and Behie, 2015; Gao, 2026). In addition, reported susceptible exposure windows have varied across studies, with some investigations identifying associations during the first trimester (Zhang, 2023; Abdo, 2019); others during the second (Holstius et al., 2012) or third trimester (Fang, 2020; Holstius et al., 2012); and still others across the entire pregnancy period (Xiao, 2018; Li, 2021). These inconsistencies indicate ongoing uncertainty regarding critical exposure windows and the effects of fire smoke exposure on fetal growth.

Compared with wildfire-focused research, relatively few studies have explicitly investigated PFS-specific PM_{2.5} and birth outcomes. Existing work has established prescribed burning as an important contributor to local and regional PM_{2.5} and a recurrent source of exposure in populated landscapes (Jaffe, 2020; Maji, 2024; Williamson et al., 2012), yet quantitative evidence linking PFS to fetal growth remains understudied. Because prescribed burning is a planned and necessary management practice, evidence on its potential health impacts is essential for developing informed public health guidance and mitigation strategies that operate alongside fire management goals. Accordingly, studies that isolate PFS-related PM_{2.5} and evaluate both trimester-specific and overall associations with birth weight are needed. In addition, because susceptibility to air pollution during pregnancy may vary by maternal and infant characteristics, examining whether associations differ across demographic and reproductive subgroups may provide further insight into potentially vulnerable populations.

In this study, we quantified prenatal exposure to PFS PM_{2.5} and evaluated its association with birth weight across multiple gestational windows in Georgia, USA. By integrating source-attributed exposure estimates with detailed birth records, this analysis addresses a critical knowledge gap at the intersection of fire management, air quality, and perinatal health. Rather than questioning the value of prescribed fire, this work seeks to inform balanced and health-protective approaches that support both the ecological benefits of fire and the well-being of exposed populations. Effective mitigation of smoke-related health impacts requires informed policy, coordinated public health action, and an informed public, particularly in regions where prescribed burning remains an essential land management tool. As one of the first epidemiologic studies to distinguish PFS-related PM_{2.5} from other PM_{2.5} sources, our findings provide new evidence to support more health-conscious prescribed fire practices alongside existing management objectives.

2. Material and methods

2.1. Study design and population

This study was a retrospective population-based cohort study of singleton term births in Georgia, USA. We linked individual birth records with prenatal exposure estimates to evaluate the association between PFS PM_{2.5} during pregnancy and birth weight. The source population included all live births with a recorded last menstrual period (LMP) date between January 1, 2013 and December 31, 2016. We focused on singleton term births, defined as births occurring at 37 to 42 completed weeks of gestation, to reduce heterogeneity related to multiple gestation and extreme variation in gestational duration, and to better evaluate differences in fetal growth within the term birth population. The final analytic cohort was established after applying pre-specified exclusion criteria, as described below.

2.2. Birth data

Birth records were obtained from the Office of Health Indicators for Planning at the Georgia Department of Public Health. The dataset contained individual-level information on infant and maternal characteristics, including LMP date, gestational age, and birth date; newborn sex (female or male); maternal age, race (White, Black, Asian/Native

Hawaiian [NH]/Pacific Islander [PI], American Indian [AI]/Alaska Native [AN], other), ethnicity (Hispanic or non-Hispanic), education (less than high school, high school/general education diploma [GED], some college/associate degree [AA], bachelor's degree or higher), marital status (married or unmarried), parity (first, second, third or higher pregnancy), reported tobacco use (yes or no) and alcohol use (yes or no) during pregnancy, and maternal residential ZIP code and county of residence at delivery.

2.3. Exposure data

Daily concentrations of total ambient PM_{2.5} and PFS PM_{2.5} were generated using a multistage air quality modeling and data-fusion approach. Detailed descriptions of the fire identification, emission allocation, atmospheric simulation, and observation fusion procedures are provided elsewhere (Maji, 2024; Maji, 2024). Briefly, prescribed burns were identified applying a fire classification algorithm to the Fire INventory from NCAR (FINN, version 2.5), and burned area was subsequently calibrated using state burn permit and activity records to mitigate the inaccuracies associated with satellite observations (Li, 2023; Wiedinmyer, 2023). Prescribed fire emissions and smoke plume heights were calculated using the BlueSky framework (Larkin, 2009). The contributions of prescribed burns to 24-hour average PM_{2.5} concentrations were then simulated with the Community Multiscale Air Quality (CMAQ; version 5.4) model driven by meteorology from the Weather Research and Forecasting (WRF) model on a 12 km × 12 km horizontal resolution grid over the southeastern United States as the difference made by prescribed fire emissions. Finally, the modeled PM_{2.5} fields were statistically fused with ground-based EPA Air Quality System (AQS) observations to reduce model bias and improve agreement with daily observations, and PFS PM_{2.5} fields were scaled accordingly. Model evaluation indicated that CMAQ alone tended to underestimate PM_{2.5} concentrations by approximately 5.8 % compared with AQS observations across twelve southeastern United States (Alabama, Arkansas, Florida, Georgia, Kentucky, Louisiana, Mississippi, North Carolina, South Carolina, Tennessee, Virginia and West Virginia) during 2013–2021. After data fusion, the bias was reduced to 1.5 % overestimation, and the model demonstrated improved performance (R^2 changed from 0.29 to 0.69) relative to EPA measurements (Maji, 2024). Final exposure surfaces were generated at a 12 km × 12 km spatial resolution and subsequently prepared for linkage to health records.

For this study, we extracted total ambient PM_{2.5} and PFS PM_{2.5} exposure data for Georgia from January 1, 2013, to December 31, 2017, to cover the entire pregnancy period for all eligible births. Note that the PM_{2.5} concentration bias in Georgia was only 1.0 % and R^2 was 0.76, both better than the rest of the domain. The original 12-km grid estimates were interpolated to the ZIP code level using an area-weighting method. Non-PFS PM_{2.5} concentrations were calculated by subtracting PFS PM_{2.5} from total ambient PM_{2.5}. These exposure estimates were then linked to birth records based on maternal residential ZIP code.

For each birth record, two sets of exposure metrics were calculated: 1) the daily average PFS PM_{2.5} concentration, and 2) the number of days with daily PFS PM_{2.5} concentrations exceeding the 75th, 90th, 95th, and 99th percentiles. Exposure metrics were assessed across multiple gestational windows, including the first 4 and 6 weeks, the first trimester (weeks 1–13), the second trimester (weeks 14–26), the third trimester (weeks 27 to delivery), and the entire pregnancy. The first 4- and 6-week windows were specifically included to capture the earliest stages of pregnancy and implantation, as recent evidence suggests that very early gestation (weeks 1–6) represents a sensitive developmental period during which PM_{2.5} exposure is associated with impaired fetal growth and reduced birth weight (Chen, 2023).

Daily maximum and minimum temperatures (Tmax and Tmin) from January 1, 2013, to December 31, 2017, were obtained from the Daymet Version 4 dataset at a 1 km spatial resolution, hosted by the Oak Ridge National Laboratory Distributed Active Archive Center (Thornton,

2021). Daily mean temperature was calculated as the arithmetic average of Tmax and Tmin. The 1 km gridded temperature data were spatially aggregated to the ZIP code level to align with the geographic scale of the birth records.

2.4. Statistical analysis

Linear regression models were used to examine the association between prenatal exposure to PFS PM_{2.5} and birth weight, with separate models specified for each gestational exposure window. Spearman's correlation analysis showed a relatively low correlation between PFS PM_{2.5} and non-PFS PM_{2.5} concentrations ($\rho = 0.29$). Therefore, both variables were included in the same model to estimate the independent effect of PFS PM_{2.5}. All models additionally were adjusted for LMP date (modeled using a natural cubic spline with 4 degrees of freedom (df) to capture long-term trends), LMP season (to capture seasonal patterns), gestational age in completed weeks (modeled as a factor variable to adjust for the fact that babies tend to gain more weight the longer they gestate), newborn sex, maternal age (modeled as a nonlinear function using a natural cubic spline with 3 df), race, ethnicity, educational attainment, marital status, parity, reported tobacco use and alcohol use during pregnancy, county of residence at delivery, and ZIP code-level poverty percentage.

The association between non-PFS PM_{2.5} and birth weight was estimated using the same regression model to facilitate comparison with the effects of PFS PM_{2.5}. Results were presented as mean differences in birth weight per interquartile range (IQR) and per 1 $\mu\text{g}/\text{m}^3$ increase in average exposure concentrations across different windows. IQRs were calculated separately for each trimester and whole-pregnancy exposures across all births. For threshold-based metrics, we reported mean changes in birth weight per additional day with PFS PM_{2.5} concentrations exceeding the specified percentile cutoffs.

We further conducted stratified analyses by gestational age, newborn sex, maternal age, race, ethnicity, education, and parity to identify potentially susceptible subgroups. Sensitivity analyses were performed to test the robustness of our findings. First, we varied the df for LMP date (3–6) and maternal age (3–6). Second, we additionally adjusted for temperature. Third, to evaluate whether variability in gestational window length influenced the corresponding estimates, we conducted an additional sensitivity analysis using fixed gestational exposure windows, in which the third trimester was restricted to weeks 27–36 and the entire pregnancy was restricted to weeks 1–36 for all births. Fourth, to assess the potential influence of area-level spatial and socioeconomic adjustment, we calculated Spearman's correlations between PFS PM_{2.5} concentrations and poverty percentage across different gestational exposure windows and fitted reduced models without poverty percentage and without county. All analyses were performed using R version 4.3.2.

3. Results

The source population included all live births with a recorded LMP between January 1, 2013 and December 31, 2016 ($n = 521,625$). We then applied a series of sequential exclusions: 1) births with missing or invalid maternal ZIP codes ($n = 486$; 0.09 %); 2) births outside 37 to 42 completed weeks of gestation ($n = 57,266$; 10.99 %); 3) births to mothers younger than 18 years or older than 45 years ($n = 8,962$; 1.93 %); 4) multiple births ($n = 6,366$; 1.40 %); and 5) births with extreme birth weights below 1,000 g or above 6,000 g (Holstius et al., 2012) ($n = 68$; 0.02 %). The final analytic cohort included 448,477 singleton term births.

Table 1 presents the descriptive statistics of the study cohort. The mean birth weight was 3342.37 g, and the mean gestational age was 38.93 weeks. Most infants were born at 39 weeks (44.09 %), followed by 40 weeks (21.39 %) and 38 weeks (18.93 %). A total of 220,425 infants were female, accounting for 49.15 % of the cohort. Overall, the mean maternal age was 28.19 years and, when grouped into age categories,

Table 1
Descriptive statistics of the birth cohort in Georgia, 2013–2017 ($n = 448,477$).

Variable	Category	Statistics
Birth weight (g)	Mean (SD)	3342.37 (461.38)
Gestational age (week)	Mean (SD)	38.93 (1.02)
	37, n (%)	45,843 (10.22)
	38, n (%)	84,877 (18.93)
	39, n (%)	197,743 (44.09)
	40, n (%)	95,914 (21.39)
	41, n (%)	22,983 (5.12)
Newborn sex	42, n (%)	1,117 (0.25)
	Female, n (%)	220,425 (49.15)
Maternal age (year)	Male, n (%)	228,052 (50.85)
	Mean (SD)	28.19 (5.69)
	≥ 18 and ≤ 25 , n (%)	158,712 (35.39)
	> 25 and ≤ 30 , n (%)	133,724 (29.82)
	> 30 and ≤ 35 , n (%)	105,404 (23.50)
Maternal race	> 35 and ≤ 45 , n (%)	50,637 (11.29)
	White, n (%)	260,749 (58.14)
	Black, n (%)	148,910 (33.20)
	Asian/NH/PI, n (%)	21,567 (4.81)
	AI/AN, n (%)	781 (0.17)
	Other, n (%)	11,173 (2.49)
	Missing, n (%)	5,297 (1.18)
Maternal ethnicity	Non-Hispanic, n (%)	381,938 (85.16)
	Hispanic, n (%)	61,415 (13.69)
	Missing, n (%)	5,124 (1.15)
Maternal education	Less than high school, n (%)	16,255 (3.63)
	High school/GED, n (%)	42,758 (9.53)
	Some college/AA, n (%)	131,382 (29.30)
	Bachelor's degree or higher, n (%)	251,000 (55.97)
	Missing, n (%)	7,072 (1.58)
Marital status	Married, n (%)	253,787 (56.59)
	Unmarried, n (%)	193,830 (43.22)
	Missing, n (%)	860 (0.19)
Parity	First pregnancy, n (%)	174,455 (38.90)
	Second pregnancy, n (%)	139,007 (31.00)
	Third or higher pregnancy, n (%)	134,994 (30.10)
	Missing, n (%)	21 (0.005)
Tobacco use during pregnancy	No, n (%)	421,529 (93.99)
	Yes, n (%)	23,855 (5.32)
	Missing, n (%)	3,093 (0.69)
Alcohol use during pregnancy	No, n (%)	373,490 (83.28)
	Yes, n (%)	980 (0.22)
	Missing, n (%)	74,007 (16.50)

Note: SD = standard deviation; NH = Native Hawaiian; PI = Pacific Islander; AI = American Indian; AN = Alaska Native; GED = general education diploma; AA = associate degree.

mothers aged 18–25 years comprised the largest proportion (35.39 %), followed by those aged 25–30 years (29.82 %), 30–35 years (23.50 %), and 35–45 years (11.29 %). More than half of mothers were White (58.13 %), about one-third were Black (33.20 %), and the majority were non-Hispanic (85.16 %). Educational attainment was relatively high: 55.97 % had a bachelor's degree or higher, 29.30 % had some college or an associate degree, and fewer than 15 % had a high school diploma, GED, or less. In terms of reproductive and behavioral characteristics, 56.59 % mothers were married at the time of pregnancy. Parity was balanced, with approximately one-third being first pregnancies, one-third second pregnancies, and one-third third or higher. Most mothers reported not using tobacco (93.99 %) or alcohol (83.28 %) during pregnancy.

Descriptive statistics of daily average ambient PM_{2.5}, non-PFS PM_{2.5}, and PFS PM_{2.5} concentrations across different gestational exposure windows are shown in Table S1. The daily mean ambient PM_{2.5} concentration was approximately 7.98 $\mu\text{g}/\text{m}^3$, with 7.19 $\mu\text{g}/\text{m}^3$ attributed to non-PFS sources and 0.79 $\mu\text{g}/\text{m}^3$ to PFS. PFS PM_{2.5} concentrations exhibited relatively higher distributions during the first 4 weeks of

gestation compared with other windows, with the 75th, 90th, 95th, and 99th percentiles being 0.99, 1.51, 1.94, and 2.67 $\mu\text{g}/\text{m}^3$, respectively. For the entire pregnancy period, the corresponding percentiles were 0.94, 1.11, 1.21, and 1.43 $\mu\text{g}/\text{m}^3$. Detailed percentile distributions across gestational windows are provided in Table S1.

Table S2 summarizes the number of days with PFS PM_{2.5} concentrations exceeding selected percentile thresholds. On average, across different gestational windows, PFS PM_{2.5} concentrations exceeded the 75th, 90th, 95th, and 99th percentiles for approximately 2.19, 0.89, 0.46, and 0.10 days, respectively.

Tables S3–S9 present the descriptive statistics of daily average PFS PM_{2.5} concentrations across gestational exposure windows, stratified by gestational age, newborn sex, maternal age, race, ethnicity, education, and parity. Overall, the distributions of PFS PM_{2.5} were generally comparable across these subgroups.

Fig. 1 illustrates the spatial distributions of PFS PM_{2.5} (left) and non-PFS PM_{2.5} concentrations (right) across Georgia during the study period. Distinct spatial patterns were observed between the two components. PFS PM_{2.5} concentrations exhibited a clear spatial gradient, with highest levels observed in southwestern Georgia. The daily average concentration within individual ZIP codes reached up to 1.22 $\mu\text{g}/\text{m}^3$ and gradually decreased toward the northern and eastern regions. In contrast, non-PFS PM_{2.5} concentrations were spatially clustered, primarily concentrated in central and northern urban areas such as the Atlanta metropolitan region. The daily average concentration within a ZIP code reached up to 9.09 $\mu\text{g}/\text{m}^3$.

Fig. S1 shows the seasonal spatial distributions of PFS PM_{2.5} (left) and non-PFS PM_{2.5} (right) concentrations across Georgia during the study period. The seasonal daily mean concentrations of PFS PM_{2.5} ranged from 0.20 to 2.06 $\mu\text{g}/\text{m}^3$, whereas non-PFS PM_{2.5} ranged from 3.90 to 10.52 $\mu\text{g}/\text{m}^3$. Both PFS and non-PFS PM_{2.5} exhibited clear seasonal patterns. PFS PM_{2.5} concentrations peaked in spring (March–May), followed by winter (December–February) and autumn (September–November), and reached the lowest level in summer (June–August), during which the maximum was only 0.51 $\mu\text{g}/\text{m}^3$. Non-PFS PM_{2.5} concentrations showed the opposite seasonal pattern and were highest in summer, with maximum daily mean concentrations up to 10.52 $\mu\text{g}/\text{m}^3$, followed by autumn, spring and winter.

Table 2 summarizes the estimated changes in birth weight associated with an IQR increase in average PFS PM_{2.5} and non-PFS PM_{2.5} concentrations across different gestational exposure windows. The IQR of non-PFS PM_{2.5} concentrations was approximately three times larger than that of PFS PM_{2.5} across all exposure windows. Both PFS and non-PFS PM_{2.5} exposures were independently and significantly associated with reductions in birth weight for certain exposure windows. After adjusting for non-PFS PM_{2.5}, each IQR increase in PFS PM_{2.5} during the first 4 weeks and each IQR increase during the first trimester were associated with decreases in birth weight of 2.03 g (95 % confidence interval [CI]: 0.52, 3.54) and 2.40 g (95 % CI: 0.06, 4.73), respectively. After adjusting for PFS PM_{2.5}, exposure to non-PFS PM_{2.5} was consistently associated with lower birth weight across gestation, except during the second trimester.

Fig. 2 presents the estimated changes in birth weight associated with a 1 $\mu\text{g}/\text{m}^3$ increase in PFS and non-PFS PM_{2.5} concentrations across different gestational windows. For an equivalent increase in concentration, the point estimates for PFS PM_{2.5} were generally greater than those for non-PFS PM_{2.5}; however, the overlapping 95 % CIs indicated that these differences were not statistically significant. The estimates for PFS PM_{2.5} were also characterized by wider CIs than those for non-PFS PM_{2.5}. Specifically, during the first 4 weeks, each 1 $\mu\text{g}/\text{m}^3$ increase in PFS PM_{2.5} was associated with a 3.22 g decrease in birth weight (95 % CI: 0.82, 5.62), compared with a 1.60 g decrease (95 % CI: 0.56, 2.64) for non-PFS PM_{2.5}. During the first trimester, each 1 $\mu\text{g}/\text{m}^3$ increase in PFS PM_{2.5} was associated with a 4.56 g decrease in birth weight (95 % CI: 0.12, 9.00), whereas the corresponding decrease for non-PFS PM_{2.5} was 2.30 g (95 % CI: 0.75, 3.84).

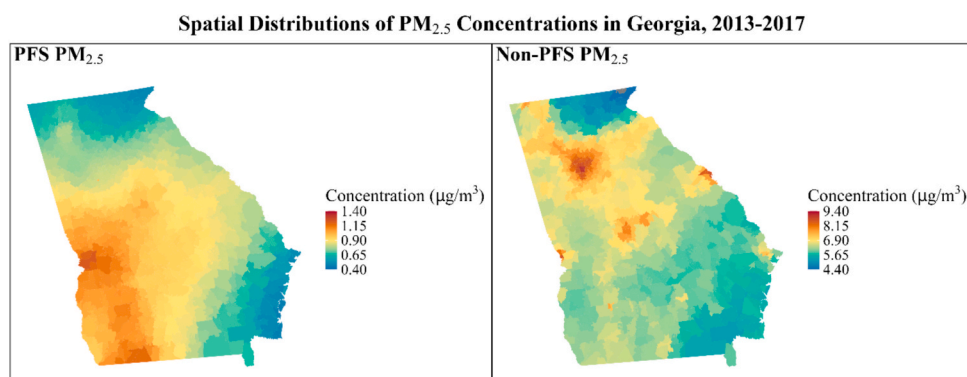


Fig. 1. Spatial distributions of PFS PM_{2.5} (left) and non-PFS PM_{2.5} concentrations (right) in Georgia, 2013–2017. Note: PFS = prescribed fire-related smoke.

Table 2

Changes in birth weight per interquartile range (IQR) increase in average exposure concentrations by gestational exposure window.

Exposure	Gestational window	IQR (µg/m ³)	Changes (g)	95 % CI
PFS PM _{2.5}	First 4 weeks	0.63	−2.03	(−3.54, −0.52) **
	First 6 weeks	0.58	−1.38	(−2.98, 0.21)
	First trimester	0.53	−2.40	(−4.73, −0.06) *
	Second trimester	0.56	0.22	(−2.22, 2.66)
	Third trimester	0.54	0.11	(−2.13, 2.35)
	Entire pregnancy	0.31	−1.57	(−5.26, 2.13)
Non-PFS PM _{2.5}	First 4 weeks	2.13	−3.40	(−5.62, −1.18) **
	First 6 weeks	2.05	−2.63	(−5.04, −0.22) *
	First trimester	1.91	−4.38	(−7.33, −1.43) **
	Second trimester	1.93	−2.45	(−5.34, 0.43)
	Third trimester	1.89	−7.11	(−9.97, −4.25) ***
	Entire pregnancy	1.51	−11.46	(−15.51, −7.41) ***

Note: IQR = interquartile range; CI = confidence interval; PFS = prescribed fire-related smoke; * indicates p-value less than 0.05; ** indicates p-value less than 0.01; *** indicates p-value less than 0.001.

Table 3 shows the estimated changes in birth weight per additional day with PFS PM_{2.5} concentrations exceeding specified percentile thresholds across different gestational exposure windows. When the threshold was set at the 75th percentile, each additional day with PFS PM_{2.5} concentrations above 1.00 µg/m³ during the first trimester was associated with a 2.48 g decrease in birth weight (95 % CI: 0.44, 4.52). At the 90th percentile (1.51 µg/m³ for the first 4 weeks and 1.46 µg/m³ for the first 6 weeks), each additional day of exceedance was associated with decreases in birth weight of 2.90 g (95 % CI: 1.25, 4.55) and 2.33 g (95 % CI: 0.40, 4.26), respectively. When the threshold was set at 95th percentile, stronger associations were observed. Each additional day exceeding the 95th percentile (1.94 µg/m³ for the first 4 weeks, 1.83 µg/m³ for the first 6 weeks, and 1.61 µg/m³ for the first trimester) was associated with decreases in birth weight of 3.76 g (95 % CI: 1.48, 6.04), 3.19 g (95 % CI: 0.47, 5.90), and 6.34 g (95 % CI: 1.93, 10.75), respectively. A significant association was also observed for the full gestation period, where each additional day exceeding the 95th percentile (1.21 µg/m³) corresponded to an 11.70 g decrease in birth weight (95 % CI: 0.95, 22.45). No significant associations were found between birth weight and the number of days with PFS PM_{2.5} concentrations exceeding the 99th percentile.

Stratified analyses (Tables S10–S23) show that the associations between PFS PM_{2.5} and reduced birth weight were detected in several

subgroups. In analyses stratified by gestational age, the association between PFS PM_{2.5} and reduced birth weight was mainly observed among births at 38–41 gestational weeks (Tables S10–S11). Associations were also observed among female infants (Tables S12–S13) and among mothers aged 18–25 years and 30–35 years (Tables S14–S15). When stratified by race, significant associations were identified among White and Asian/NH/PI mothers (Tables S16–S17). Among maternal ethnicity groups, significant associations were detected among non-Hispanic mothers for metrics based on days with PFS PM_{2.5} concentrations exceeding 90th, 95th, and 99th percentiles (Tables S18–S19). By maternal education level, associations were observed among mothers with less than a high school education or with a high school or GED diploma (Tables S20–S21). Finally, parity-stratified analyses suggested that associations were present among mothers in their first or third pregnancy (Tables S22–S23). Overall, these findings suggest possible heterogeneity in susceptibility across maternal and infant subgroups.

Tables S24–S31 and S33–36 show that the results of sensitivity analyses were generally consistent with those from the main models. Varying the df for LMP date and maternal age, additionally adjusting for temperature, applying fixed gestational exposure windows, and removing poverty percentage or county from the models did not alter the overall pattern of associations. Spearman's correlations between PFS PM_{2.5} concentrations and poverty percentage were low across gestational exposure windows, with coefficients ranging from 0.03 to 0.09 (Table S32).

4. Discussion

In this retrospective cohort study conducted in Georgia, USA, we found that both PFS PM_{2.5} and non-PFS PM_{2.5} were independently associated with reductions in birth weight after mutual adjustment. Although the point estimates per 1 µg/m³ increase were larger for PFS PM_{2.5} than for non-PFS PM_{2.5} in some gestational windows, these differences were not statistically significant. This finding should be interpreted cautiously and regarded as hypothesis-generating rather than conclusive evidence of greater toxicity per unit mass for PFS PM_{2.5}.

Our findings for non-PFS PM_{2.5} are broadly consistent with the extensive literature linking maternal exposure to ambient PM_{2.5} and reduced birth weight. In the existing epidemiologic literature, ambient PM_{2.5} typically refers to total outdoor PM_{2.5} from all sources and has been repeatedly associated with lower birth weight in multi-center studies and systematic reviews (Dadvand, 2013; Ghosh, 2021; Gong et al., 2022; Individually-varied susceptibility and spatial heterogeneity, 2018). In our study, non-PFS PM_{2.5} was defined as total ambient PM_{2.5} minus PFS PM_{2.5} and therefore represents a mixture of PM_{2.5} from non-prescribed fire sources, which may include background ambient sources and, to a lesser extent, WFS. Because no prior studies have explicitly examined non-PFS PM_{2.5} as defined here, we compare our estimates with published results for ambient PM_{2.5}.

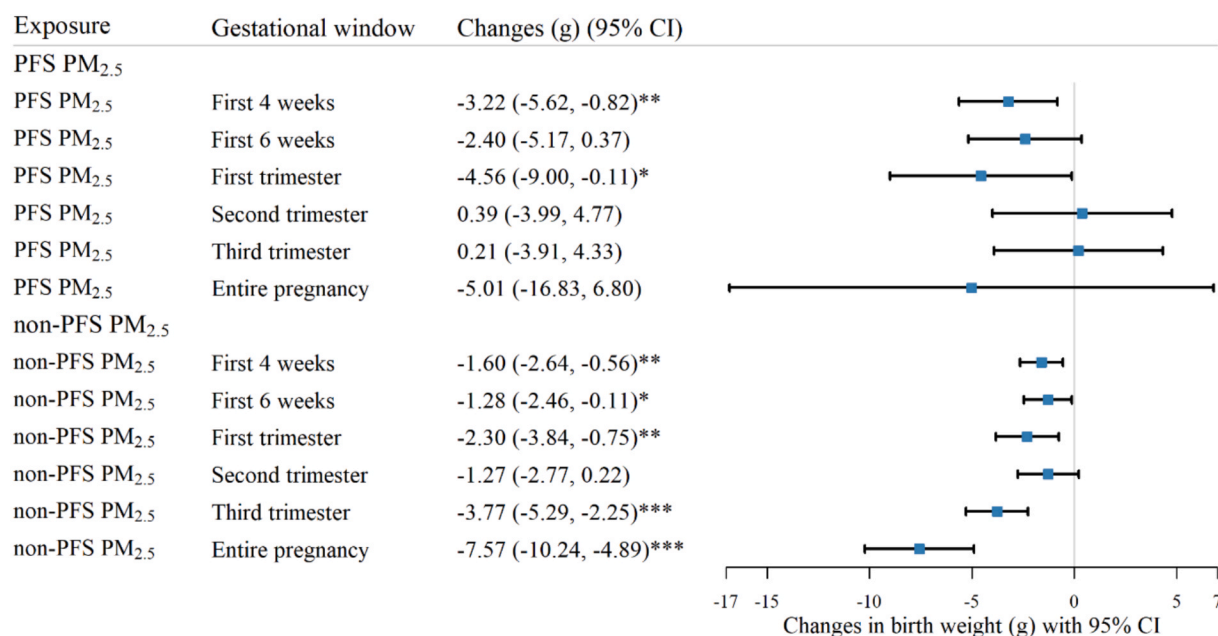


Fig. 2. Changes in birth weight per 1 $\mu\text{g}/\text{m}^3$ increase in average exposure concentrations by gestational exposure window. Note: CI = confidence interval; PFS = prescribed fire-related smoke; * indicates p-value less than 0.05; ** indicates p-value less than 0.01; *** indicates p-value less than 0.001.

Table 3

Changes in birth weight per additional day with PFS PM_{2.5} concentrations exceeding specified percentiles by gestational exposure window.

Percentile cutoff	Gestational window	Changes (g)	95 % CI
75th	First 4 weeks	-1.16	(-2.35, 0.03)
	First 6 weeks	-1.16	(-2.51, 0.20)
	First trimester	-2.48	(-4.52, -0.44) *
	Second trimester	1.65	(-0.40, 3.70)
	Third trimester	0.51	(-1.49, 2.52)
	Entire pregnancy	-0.04	(-5.43, 5.36)
90th	First 4 weeks	-2.90	(-4.55, -1.25) ***
	First 6 weeks	-2.33	(-4.26, -0.40) *
	First trimester	-2.84	(-5.79, 0.11)
	Second trimester	0.61	(-2.31, 3.53)
	Third trimester	-0.36	(-3.12, 2.41)
	Entire pregnancy	-1.16	(-8.49, 6.18)
95th	First 4 weeks	-3.76	(-6.04, -1.48) **
	First 6 weeks	-3.19	(-5.90, -0.47) *
	First trimester	-6.34	(-10.75, -1.93) **
	Second trimester	-1.33	(-5.72, 3.07)
	Third trimester	-1.48	(-5.63, 2.66)
	Entire pregnancy	-11.70	(-22.45, -0.95) *
99th	First 4 weeks	-4.52	(-10.14, 1.11)
	First 6 weeks	-2.69	(-9.26, 3.89)
	First trimester	-6.06	(-16.28, 4.16)
	Second trimester	-5.72	(-15.75, 4.30)
	Third trimester	-2.32	(-11.89, 7.25)
	Entire pregnancy	-19.79	(-43.00, 3.43)

Note: PFS = prescribed fire-related smoke; CI = confidence interval; * indicates p-value less than 0.05; ** indicates p-value less than 0.01; *** indicates p-value less than 0.001.

The magnitude of birth weight reduction associated with non-PFS PM_{2.5} in our study was larger than several pooled estimates for ambient PM_{2.5}. For example, a global *meta-analysis* found an overall decrease of 2.2 g in birth weight per 1 $\mu\text{g}/\text{m}^3$ increase in ambient PM_{2.5} during the entire pregnancy (Ghosh, 2021); whereas our study estimated a 7.57 g decrease per 1 $\mu\text{g}/\text{m}^3$ increase in non-PFS PM_{2.5}. The differences may reflect variation in population characteristics, regional susceptibility in the southeastern United States, exposure assessment approaches, or covariate adjustment. A Finnish cohort study reported a comparable magnitude of association, with a 6.5 g decrease in birth

weight per 1 $\mu\text{g}/\text{m}^3$ increase in ambient PM_{2.5} during the entire pregnancy (Rumrich, 2025). In addition, another investigation observed stronger effects at lower ambient PM_{2.5} concentrations, with birth weight declining sharply up to 10.74 $\mu\text{g}/\text{m}^3$ and plateauing thereafter (Fu, 2022). The 99th percentile of non-PFS PM_{2.5} in our study (9.41 $\mu\text{g}/\text{m}^3$ during the entire pregnancy; Table S1) lies within the steeper portion of that exposure-response curve, which may partly explain the larger effect estimates observed at relatively lower concentrations.

Few studies have directly assessed the relationship between PFS PM_{2.5} and birth weight. Previous research on prenatal exposure to WFS PM_{2.5} and birth weight has reported inconsistent findings, with some studies observing higher birth weight (Fernandez, 2023; O'Donnell and Behie, 2015; Gao, 2026) and others reporting lower birth weight (Zhang, 2023; Abdo, 2019; Li, 2021). Consistent with our estimate of a 4.56 g (95 % CI: 0.11, 9.00) decrease per 1 $\mu\text{g}/\text{m}^3$ increase in PFS PM_{2.5} during the first trimester, a study conducted in Colorado, USA reported a comparable 5.7 g (95 % CI: 0.4, 11.1) decrease associated with WFS PM_{2.5} (Abdo, 2019). Although several estimated reductions in birth weight were statistically significant, their absolute magnitudes were small, and the clinical significance of a reduction of a few grams for an individual infant is uncertain. Nevertheless, modest shifts in birth weight may still be relevant from a population-level public health perspective in regions where recurrent PFS exposure affects large numbers of pregnancies. Against this background, our findings provide new evidence on the association between PFS PM_{2.5} exposure and reduced birth weight.

Several factors may have contributed to the observed pattern of larger point estimates for PFS PM_{2.5}. The CIs for PFS PM_{2.5} were wider, indicating lower precision. In addition, the exposure distributions of PFS PM_{2.5} and non-PFS PM_{2.5} differed substantially in our study population. The relatively low concentration range of PFS PM_{2.5} may have contributed to the apparent difference in per-unit estimates and may limit direct comparison with non-PFS PM_{2.5}, as comparisons based on a 1 $\mu\text{g}/\text{m}^3$ increment may be inherently asymmetric. Potential nonlinearity in the exposure-response relationship may also influence the comparability of per-unit estimates. Therefore, the observed differences in point estimates should be regarded as hypothesis-generating and should not be interpreted as evidence of greater toxicity per unit mass for PFS PM_{2.5}.

Although our study did not measure particle composition or oxidative potential, prior literature on fire smoke provides possible biologic context for the observed associations (Aguilera et al., 2021). While prescribed fires and wildfires differ in combustion conditions and overall smoke composition, insights from the wildfire literature remain informative for understanding potential mechanisms. Studies of WFS indicate that, compared with non-smoke PM_{2.5}, smoke-derived PM_{2.5} contains a higher proportion of ultrafine particles and compounds with greater oxidative potential, which can more effectively trigger cellular oxidative stress (Stowell, 2019). Fire smoke also contains elevated concentrations of combustion-related constituents such as organic carbon, black carbon, and PAHs (Wegesser et al., 2010); all of which have been implicated in adverse developmental outcomes. Prescribed fires generally burn under cooler and moister conditions that favor smoldering combustion, producing PM_{2.5} enriched in organic carbon and semi-volatile organic compounds relative to non-smoke sources (Jaffe, 2020; Kelp, 2023). These smoldering-related emissions may provide biologic plausibility for associations between PFS PM_{2.5} and fetal growth, but they should not be taken as direct evidence of greater toxicity of PFS PM_{2.5} in this study.

We also identified different sensitive windows for PFS PM_{2.5} and non-PFS PM_{2.5}. PFS PM_{2.5} showed the strongest associations during very early pregnancy and the first trimester, whereas non-PFS PM_{2.5} was associated with reduced birth weight across multiple gestational windows, though not consistently during the second trimester. Heightened sensitivity to PFS PM_{2.5} during early gestation is biologically plausible. The first 4- and 6-weeks of pregnancy encompass implantation, trophoblast invasion, early placental development, and utero-placental vascular formation (Donovan and Cascella, 2026). These processes are critical for establishing effective maternal-fetal circulation and have been mechanistically linked to subsequent fetal growth and birth weight (Burton and Fowden, 2015; Woods et al., 2018). Impaired placental development and inadequate spiral artery remodeling can lead to fetal growth restriction (Knofler, 2019). If PFS PM_{2.5} disrupts trophoblast function, angiogenesis, or oxidative balance during these formative stages, nutrient and oxygen transport to the fetus may be compromised. Supporting this interpretation, another cohort study has demonstrated that uteroplacental vascular development during the first trimester is critical for fetal growth (Reijnders, 2021). Differences in timing of susceptibility between PFS and non-PFS PM_{2.5} may result from variations in particle composition, co-pollutants, and exposure patterns. Smoke events are often episodic, characterized by short-term peaks with elevated organic carbon, black carbon, and combustion-specific compounds such as PAHs, which may exert acute biological effects during critical windows of placental development.

We further examined the effects of different exposure metrics and found that they influenced the identified sensitive windows. When average concentration metrics were used, associations were most evident during early pregnancy and the first trimester. In contrast, when exposure was defined by the number of days exceeding a high percentile threshold of PFS PM_{2.5} (e.g., the 90th percentile), associations extended across the entire pregnancy. This difference may suggest that intensity and frequency of high-smoke days capture additional features of exposure beyond long-term averages. Intense smoke episodes may deliver acute doses that disrupt placental processes or trigger systemic maternal responses, even when overall mean concentrations remain moderate. A recent WFS study similarly emphasized the importance of incorporating both average exposure and high-event metrics when evaluating health effects of smoke-related PM_{2.5} (Casey, 2024).

However, although associations were observed for the 95th percentile metric, the corresponding estimates for the 99th percentile were uniformly non-significant, and this pattern should be interpreted cautiously. It may partly reflect reduced statistical power, as days exceeding the 99th percentile were much less frequent than days exceeding the 95th percentile across gestational exposure windows. At the same time, a nonlinear or plateau-type relationship at the highest

exposure levels cannot be excluded and should be explored in future studies. More broadly, our findings based on both average exposure metrics and percentile-based high-exposure indicators may further motivate future studies using flexible modeling approaches, such as splines, to characterize potential nonlinear exposure-response relationships across the PFS PM_{2.5} exposure distribution.

Our stratified analyses also suggested that associations between PFS PM_{2.5} and birth weight may vary across maternal and infant subgroups. More evident associations were observed in some gestational age, infant sex, maternal age, race/ethnicity, education, and parity strata, suggesting possible heterogeneity in susceptibility. These differences may reflect variation in biologic vulnerability, baseline health status, social and environmental conditions, or exposure patterns. However, these subgroup-specific findings should be interpreted cautiously, as the stratified analyses were exploratory, involved multiple comparisons across exposure windows, metrics, and strata, and some subgroup estimates were imprecise because of limited sample size. Accordingly, these results should be viewed as hypothesis-generating rather than as definitive evidence of effect modification. Further studies are needed to confirm whether certain maternal or infant populations are more susceptible to PFS exposure during pregnancy.

This study is among the first to distinguish PFS PM_{2.5} from other PM_{2.5} sources and evaluate its association with birth weight in the southeastern United States. Nonetheless, several limitations should be acknowledged. First, some degree of exposure assessment error is unavoidable. Although our classification approach was designed to isolate PFS, misclassification of fire sources may still occur. For example, a limitation of the FINN-based algorithm is that if a wildfire is extinguished on the same day it is detected, it may be classified as a prescribed fire; consequently, a portion of emissions labeled as PFS may originate from WFS (Maji, 2024). Additionally, satellite-based fire emissions inventories such as FINN may miss small burns, fires occurring outside the satellite overpass time, or fires obscured by cloud cover, leading to underestimation of exposure (Stowell, 2025). The CMAQ-based exposure estimates were generated at a 12 km × 12 km resolution, which may not fully capture fine-scale spatial variability in smoke exposure. Furthermore, residential address at delivery may not fully represent maternal mobility throughout pregnancy, and grid-based exposure estimates cannot capture individual-level variability in exposure related to time-activity patterns, indoor infiltration, air filtration use, occupational exposures, and other behaviors that may influence personal exposure. Such exposure measurement error may have contributed to uncertainty in the estimated associations. It may also have differed across gestational windows, particularly because shorter windows and episodic smoke exposures may be more sensitive to spatial and temporal misclassification than longer averaging periods. In addition, the prescribed burning source category in this study mainly reflects managed forest burning within the emissions modeling framework and may not fully capture other types of open burning, such as agricultural fire, which may limit the scope of interpretation of the findings.

Second, this study focused on a single state, which may limit the generalizability of our findings to other regions with different burning practices, fuel types, meteorological conditions, and population susceptibility. Third, as with all observational epidemiology, residual confounding cannot be completely ruled out, and residual spatial confounding remains an important limitation. Although our models adjusted for county, poverty percentage, LMP season, and long-term temporal trends, and sensitivity analyses additionally adjusted for temperature, these covariates may not have fully captured all regional differences associated with higher PFS PM_{2.5} levels in southwestern Georgia. In sensitivity analyses, the correlations between PFS PM_{2.5} concentrations and poverty percentage were weak, and the estimated associations were generally similar in models without poverty percentage or county. However, low correlations with a single socioeconomic indicator do not fully address the broader issue of spatial confounding, which may arise from multiple geographically patterned factors beyond

poverty. Other unmeasured regional co-exposures or contextual factors correlated with prescribed burning activity, including healthcare access, regional stressors, agricultural activity, housing characteristics, occupational exposures, smoking prevalence, broader socioeconomic conditions, and other environmental exposures, may still have contributed to the observed associations.

We also note that the third trimester was defined as weeks 27 to delivery in the main analysis, resulting in some variability in exposure window length across births. Although gestational age was adjusted for in the main models to reduce potential bias related to differences in pregnancy duration, heterogeneity in the third-trimester exposure window could still have influenced the corresponding estimates. Results from the fixed-window sensitivity analysis were generally similar to those from the main analysis, suggesting that the findings were not materially driven by variability in exposure-window length. This sensitivity analysis addressed exposure-window standardization and should be distinguished from the potential causal role of gestational age. Separately, because the analysis was restricted to singleton term births and adjusted for gestational age, our estimated associations should be interpreted as associations between prenatal PFS PM_{2.5} exposure and birth weight among pregnancies that reached term. Gestational age may lie on the causal pathway between prenatal smoke exposure and birth weight; therefore, adjustment for gestational age may partially block the total effect of exposure on birth weight. Similarly, restricting the analysis to term births may introduce selection bias if exposure influences the probability of preterm delivery. As a result, our findings should not be interpreted as the total effect of prenatal smoke exposure on birth weight in the full birth population. Future studies incorporating more refined exposure characterization, richer individual- and area-level covariate information, and broader birth outcome definitions are warranted to further clarify these associations.

5. Conclusion

This statewide study found that prenatal exposure to both PFS PM_{2.5} and non-PFS PM_{2.5} was independently associated with reduced birth weight. Significant associations were identified for PFS PM_{2.5} during early pregnancy and the first trimester, highlighting early gestation as a vulnerable period. Incorporating both average exposure metrics and indicators of high-smoke days provided a more comprehensive understanding of the exposure–response relationship. These findings underscore the importance of fine-scale, source-specific exposure assessment in perinatal air pollution research and provide new evidence that exposure to PFS PM_{2.5} may adversely affect fetal growth.

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CRediT authorship contribution statement

Lei Li: Writing – review & editing, Writing – original draft, Visualization, Validation, Software, Methodology, Investigation, Formal analysis, Data curation, Conceptualization. **Hua Hao:** Writing – review & editing, Data curation. **Jennifer D. Stowell:** Writing – review & editing. **Emily M. Gargulinski:** Writing – review & editing. **Kamal J. Maji:** Writing – review & editing, Data curation. **Colleen E. Reid:** Writing – review & editing. **Talat M. Odman:** Writing – review & editing, Resources, Data curation. **Howard H. Chang:** Writing – review & editing, Resources, Data curation. **Amber Soja:** Writing – review & editing, Resources, Data curation. **Yang Liu:** Writing – review & editing, Supervision, Resources, Project administration, Funding acquisition, Conceptualization.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Appendix A. Supplementary data

Supplementary data to this article can be found online at <https://doi.org/10.1016/j.envint.2026.110379>.

Data availability

Data will be made available on request.

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